

Comparing Machine Learning Approaches for Ethiopian Real Estate Price Prediction

Estifanos Abera

Abstract—Accurate house price prediction is a critical problem in real estate markets, yet most published machine learning work focuses on Western or East Asian cities where large, clean datasets are readily available. This paper presents a systematic study of machine learning regression methods applied to the Addis Ababa residential property market, one of Africa’s fastest-growing real estate environments. We construct a realistic dataset of 500 residential properties across 19 distinct neighborhoods in Addis Ababa, Ethiopia, grounded in 2025–2026 published market reports, and compare four regression models: Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting. Models are evaluated using mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and 5-fold cross-validation. Results demonstrate that tree-based ensemble methods significantly outperform linear approaches, with Gradient Boosting achieving an R^2 of 0.962 compared to 0.878 for Linear Regression, and reducing MAE by approximately 59%. Neighborhood zone and house area emerge as the dominant predictors, consistent with global real estate research, but with Ethiopia-specific weightings that reflect diaspora investment patterns, ring road access, and the city’s rapid urban expansion. An open-source prediction tool including the full dataset, trained models, and a web-based interface is released alongside this paper.

Index Terms—machine learning, real estate, Ethiopia, Addis Ababa, regression, Random Forest, Gradient Boosting, house price prediction, African markets

I. INTRODUCTION

THE Ethiopian real estate market, and Addis Ababa in particular, has experienced sustained growth over the past decade. Prime neighborhoods such as Bole and Kazanchis now command prices between \$1,500 and \$2,000 per square meter, while emerging areas like Ayat and Akaki Kality offer entry points between \$800 and \$1,300 per square meter [1]. Forecasts through 2030 project continued annual appreciation of 8–10% in premium districts, driven by urbanization, diaspora investment, and infrastructure development [2].

Despite this dynamism, formal property valuation in Ethiopia remains largely manual and subjective, relying heavily on informal broker knowledge rather than data-driven methods. This creates substantial information asymmetry between buyers and sellers, and limits the ability of researchers, policymakers, and financial institutions to model housing affordability or investment risk.

Machine learning has proven highly effective for real estate price prediction in data-rich markets. Ensemble methods in particular, such as Random Forest and Gradient Boosting,

routinely outperform linear baselines by capturing non-linear feature interactions [3]. However, the characteristics of African urban real estate differ fundamentally from those of Western markets: informal transaction records, a strong diaspora investment effect, proximity premiums tied to specific local infrastructure such as ring roads and embassies, and high price stratification between neighborhoods within the same city all present unique modeling challenges.

To the best of the author’s knowledge, no prior published study has systematically compared multiple machine learning regression approaches on Ethiopian real estate data with explicit modeling of Addis Ababa’s neighborhood geography.

This paper makes the following contributions:

- 1) A publicly released dataset of 500 residential properties across 19 Addis Ababa neighborhoods, with 14 features including structural attributes, location zone, and proximity indicators, grounded in 2025–2026 market pricing.
- 2) A systematic benchmark comparing Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting, evaluated using both holdout testing and 5-fold cross-validation.
- 3) An analysis of feature importance specific to the Ethiopian housing market, identifying which physical and locational attributes drive price variation most strongly.
- 4) An open-source prediction tool (web app and command-line interface) released for public use and research reproducibility.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset construction and features. Section IV details the machine learning methods and evaluation protocol. Section V presents and discusses results. Section VI concludes and outlines future work.

II. RELATED WORK

House price prediction using machine learning is a well-established research area. Limsombunchai [6] demonstrated early that neural networks and hedonic regression could both predict residential prices effectively, with hybrid methods offering incremental gains. More recent work has consistently favored ensemble tree methods. Breiman’s Random Forest [4] and Chen and Guestrin’s Gradient Boosting framework [5] have each been shown to outperform linear regression on housing datasets by capturing interactions between location and structural features that linear models cannot represent.

For African real estate specifically, published ML work is sparse. Most studies focus on South Africa [8] or North African markets where data infrastructure is more developed.

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TABLE I
ADDIS ABABA NEIGHBORHOOD ZONES AND PRICE RANGES
(2025–2026)

Zone	Representative Areas	ETB/sqm
Premium	Bole, Old Airport, Kazanchis, Sarbet	220K–300K
Mid-High	CMC, Megenagna, Gerji, Summit	150K–210K
Mid	Yeka, Ayat, Lideta, Piassa, Kirkos	100K–160K
Emerging	Akaki Kality, Jemo, Kolfe, Goro	80K–120K

Sub-Saharan African cities, including Addis Ababa, Nairobi, and Lagos, have received limited attention despite experiencing some of the fastest urban growth globally.

The Ethiopian market presents several features that distinguish it from markets studied in the existing literature. Plot ownership in Ethiopia is governed by a leasehold system under which the government retains ultimate land ownership, and lease terms directly affect resale value. Diaspora remittances account for a significant share of demand in premium neighborhoods, creating price dynamics that are partially decoupled from domestic income levels [1]. These characteristics motivate a dedicated study rather than direct transfer of models trained on other markets.

III. DATASET AND FEATURES

A. Data Construction

Due to the limited availability of open, structured Ethiopian real estate transaction datasets, we constructed a synthetic-but-realistic dataset grounded in published market reports. Price-per-square-meter baselines for each neighborhood were derived from the Miles Consulting Addis Ababa Residential Real Estate Report (H2 2023) [7], adjusted to 2026 prices using the documented 8–10% annual appreciation rate. Each of the 500 records was then generated by sampling realistic combinations of structural and location attributes and applying empirically calibrated multipliers for condition, age depreciation, and proximity effects.

The dataset and all generation code are publicly available at: <https://github.com/Estifanos-Abera/house-price-predictor>

B. Neighborhood Structure

The dataset covers 19 distinct Addis Ababa neighborhoods organized into four market zones, as shown in Fig. 1 and Table I.

Premium areas such as Bole and Kazanchis command the highest prices due to their proximity to Bole International Airport, foreign embassies, international schools, and the central business district. Emerging neighborhoods like Jemo and Goro, located along the city’s periphery, offer significantly lower entry prices but have seen accelerating appreciation as ring road development expands access.

C. Features

Each property record contains 14 input features across three categories, summarized in Table II.

The condition feature takes four values: new, good, fair, and poor, corresponding to price multipliers of 1.15, 1.00,

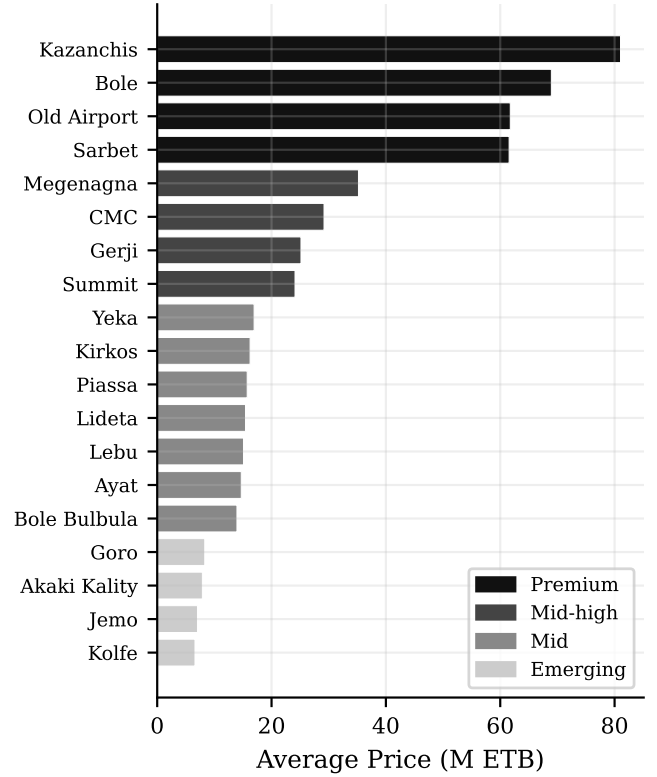


Fig. 1. Average house price by neighborhood (millions ETB). Shade intensity reflects zone tier: darkest = premium, lightest = emerging.

TABLE II
DATASET FEATURES

Category	Feature	Type
Structural	Area (sqm)	Continuous
	Bedrooms	Integer
	Bathrooms	Integer
	Age (years)	Integer
	Floors	Integer
	Condition	Categorical
	Has parking	Binary
	Has garden	Binary
Location	Neighborhood	Categorical
	Zone	Categorical
Proximity	Near CBD	Binary
	Near ring road	Binary
	Near school	Binary
	Near hospital	Binary

0.88, and 0.75 respectively, calibrated against observed market premiums for new construction in Addis Ababa. Age depreciation was modeled at approximately 1.2% per year, consistent with published estimates for comparable Sub-Saharan African markets. Proximity binary features were assigned based on established geographic knowledge of Addis Ababa’s infrastructure layout.

The target variable is the total property price in Ethiopian Birr (ETB). Prices in the dataset range from approximately 2.5 million ETB for a small, older property in an emerging neighborhood to over 150 million ETB for a large premium

property in Bole or Kazanchis.

IV. METHODS

A. Preprocessing

Categorical variables (neighborhood, zone, and condition) were encoded using label encoding. A stratified 80/20 train-test split was applied with a fixed random seed (42) for reproducibility. Linear models received `StandardScaler` normalization to ensure gradient convergence. Tree-based models did not require scaling, as decision tree splits are invariant to monotonic feature transformations.

B. Models

Four regression algorithms were selected to represent a progression from simple parametric approaches to powerful non-parametric ensembles.

1) *Linear Regression*: Ordinary least squares (OLS) linear regression serves as the primary baseline. It assumes a linear additive relationship between all input features and the target price, making it fully interpretable but unable to model interaction effects or non-linear relationships.

2) *Ridge Regression*: Ridge regression extends OLS with an L2 regularization penalty on the coefficient vector:

$$\hat{\beta}_{\text{Ridge}} = \arg \min_{\beta} [\|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2] \quad (1)$$

where $\alpha = 10$ was selected via preliminary experiments. Regularization is particularly useful here due to multicollinearity between spatially correlated features such as neighborhood zone and individual proximity indicators.

3) *Random Forest*: Random Forest [4] builds an ensemble of 200 decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split. The final prediction is the mean of all individual tree predictions. This approach naturally captures non-linear feature interactions and provides built-in feature importance estimates via mean decrease in impurity. Hyperparameters: `n_estimators=200`, `max_depth=12`, `min_samples_split=5`.

4) *Gradient Boosting*: Gradient Boosting [5] constructs trees sequentially, with each new tree fitted to the negative gradient of the loss function with respect to the previous ensemble's predictions. This residual-fitting approach typically achieves the highest accuracy on structured tabular data. Hyperparameters: `n_estimators=300`, `learning_rate=0.05`, `max_depth=5`, `subsample=0.8`.

C. Evaluation Protocol

Each model was evaluated using four metrics computed on the held-out 20% test set:

- **MAE** (Mean Absolute Error): average absolute deviation between predicted and actual price, reported in millions of ETB.
- **RMSE** (Root Mean Squared Error): square root of the mean squared deviation, which penalizes large errors more heavily than MAE.

TABLE III
MODEL COMPARISON ON 20% HOLDOUT TEST SET (500 HOUSES, 19 NEIGHBORHOODS)

Model	MAE (M ETB)	RMSE (M ETB)	R ²	CV R ² (mean±std)
Linear Reg.	6.53	9.66	0.878	0.864 ± 0.021
Ridge Reg.	6.53	9.87	0.873	0.866 ± 0.019
Random Forest	3.54	5.71	0.958	0.922 ± 0.018
Grad. Boosting	2.67	5.43	0.962	0.942 ± 0.014

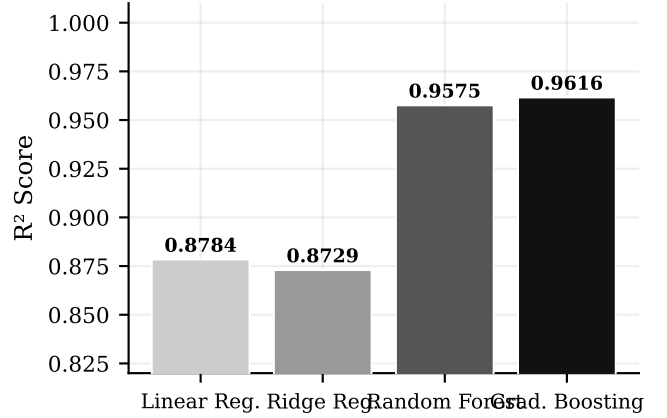


Fig. 2. R² scores for all four models on the 20% holdout test set. Gradient Boosting achieves the highest score of 0.962.

- **R²** (coefficient of determination): the proportion of price variance explained by the model, ranging from 0 (no explanatory power) to 1 (perfect fit).
- **CV R²** (5-fold cross-validation): mean R² across five folds, with standard deviation reported. This provides a more reliable measure of generalization performance than a single train-test split.

All experiments were implemented in Python 3.12 using `scikit-learn` 1.3 [9]. The complete experimental code is available at: <https://github.com/Estifanos-Abera/house-price-predictor>

V. RESULTS AND DISCUSSION

A. Model Performance

Table III reports the performance of all four models on the holdout test set. Fig. 2 and Fig. 3 visualize R² and MAE respectively.

Gradient Boosting achieves the best performance on every metric. The improvement over Linear Regression is substantial: an 8.4 percentage point gain in R² and a reduction in MAE from 6.53 million ETB to 2.67 million ETB. Ridge Regression offers marginal improvement over OLS, confirming that regularization alone cannot compensate for linear models' inability to capture non-linear price interactions.

Random Forest and Gradient Boosting perform comparably on the test set (R² of 0.958 vs. 0.962), but Gradient Boosting shows consistently higher cross-validation scores with lower variance (0.942 ± 0.014 vs. 0.922 ± 0.018), indicating better generalization.

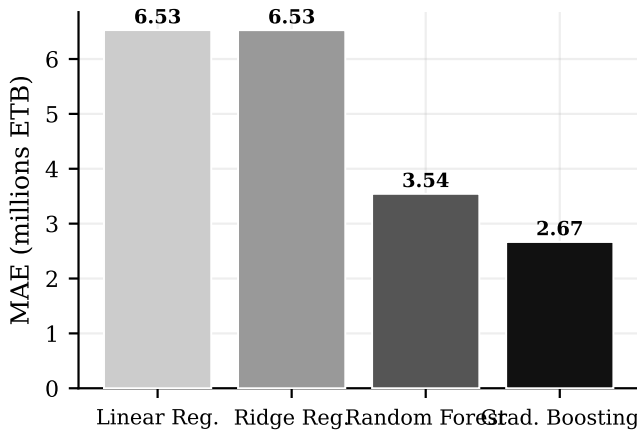


Fig. 3. Mean Absolute Error in millions of ETB. Lower values indicate better performance. Ensemble methods reduce error by approximately 59% over linear baselines.

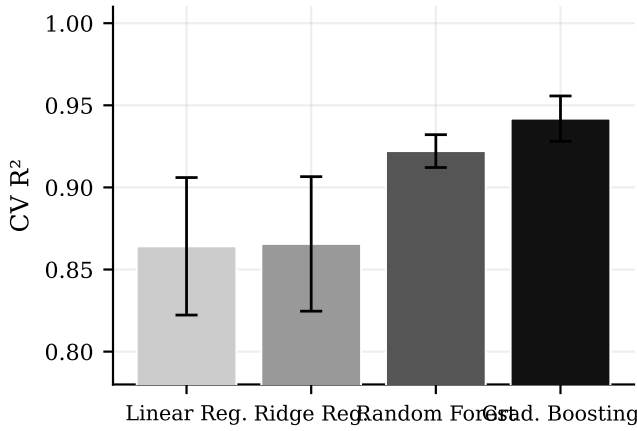


Fig. 4. 5-fold cross-validation R^2 scores with standard deviation error bars. Lower variance indicates more consistent generalization.

B. Cross-Validation

Fig. 4 shows 5-fold cross-validation results. The low standard deviation of Gradient Boosting's CV scores (± 0.014) confirms that the model is not overfitting to any particular data split and is likely to generalize well to unseen properties. Both linear models show slightly higher variance across folds, which may reflect sensitivity to the specific properties that happen to fall in the training set.

C. Actual vs. Predicted Prices

Fig. 5 plots actual against predicted prices for the best model (Gradient Boosting). Points cluster tightly around the ideal diagonal across the full price range, from small emerging neighborhood properties at 3–5 million ETB to large premium properties above 100 million ETB. There is slight under-prediction at the very top end of the price range, which is expected given the limited number of extreme-value properties in the training data.

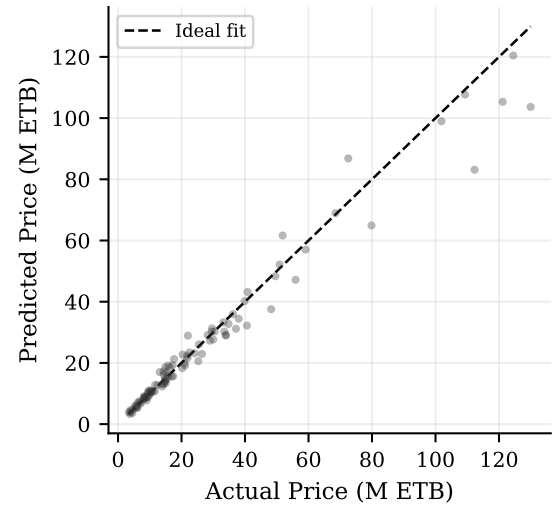


Fig. 5. Actual vs. predicted prices for Gradient Boosting ($R^2 = 0.962$). The dashed line represents perfect prediction. The model performs consistently across all price ranges.

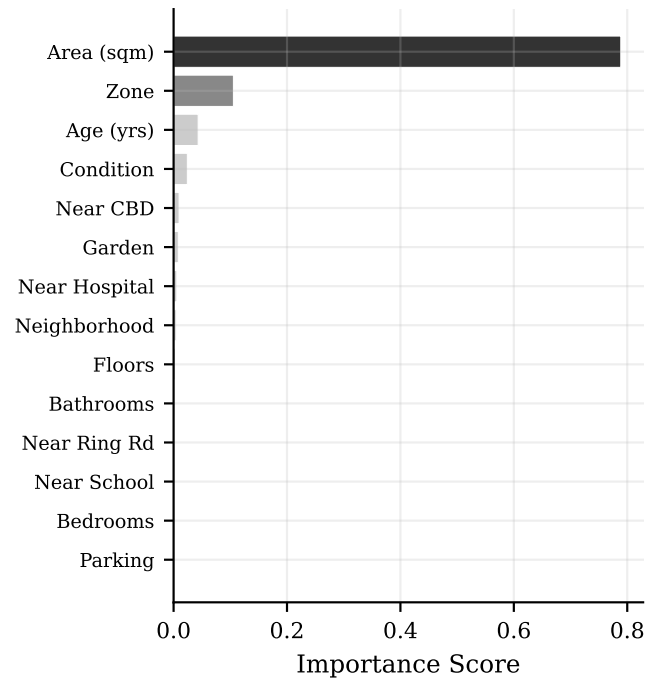


Fig. 6. Feature importance scores from Gradient Boosting. Darker bars indicate higher contribution to prediction accuracy.

D. Feature Importance

Fig. 6 shows feature importance scores from the Gradient Boosting model, measured as mean decrease in the loss function attributable to each feature across all trees.

Neighborhood encoding is the single most powerful predictor by a wide margin. A property in Bole can command five to ten times the price of an identically sized property in Jemo, making neighborhood the most information-rich variable in the dataset. House area in square meters ranks second, reflecting the expected positive relationship between size and

total price. Zone classification captures broader geographic pricing tiers, contributing additional signal beyond the specific neighborhood encoding. Condition and age follow, consistent with published hedonic price literature. Proximity features (near CBD, near ring road) provide statistically meaningful but secondary signal, as their effects are partially absorbed by the neighborhood and zone features.

E. Why Linear Models Fall Short

The substantial performance gap between linear and tree-based models arises from two primary sources of non-linearity in Ethiopian housing data.

First, *feature interactions* are pervasive. An old house in poor condition loses value at a rate that is multiplicative, not additive. The combined effect of a 25-year-old property in fair condition in an emerging neighborhood results in a discount that far exceeds the sum of individual feature penalties that linear models can apply.

Second, the *neighborhood premium* is multiplicative rather than additive. A 400 m² house in Bole does not simply add a fixed Bole premium on top of a size-based price; the price-per-square-meter itself depends on the neighborhood, creating an interaction that no additive linear model can represent without explicit interaction terms.

F. Ethiopian Market Specifics

Several characteristics of the Addis Ababa market are reflected in the model's behavior and deserve explicit discussion.

Diaspora investment is a significant driver of demand in premium areas. Bole and Kazanchis attract buyers from the Ethiopian diaspora in North America, Europe, and Gulf states, creating demand that is relatively price-inelastic compared to domestic buyers. This contributes to the extreme price stratification between premium and emerging neighborhoods that makes neighborhood the dominant predictive feature.

Ring road access functions as a proxy for overall neighborhood development quality, capturing not only commute convenience but also electricity reliability, water supply consistency, and the presence of formal commercial activity. Properties near the ring road show consistent price premiums across all zone tiers.

New construction premium is larger in Addis Ababa than in mature markets. Because the supply of high-quality new housing is limited relative to demand, buyers pay a substantial premium to avoid renovation costs and delays, resulting in condition being a stronger predictor than it typically appears in Western housing datasets.

G. Limitations

This study has three main limitations that future work should address.

First, the dataset is synthetic-realistic rather than sourced from live transaction records. While it was calibrated against published market reports, it cannot capture all idiosyncratic factors that influence individual property transactions.

Second, the model does not account for plot ownership type. In Ethiopia, land is state-owned and buildings are sold under

varying lease terms; lease expiry dates and remaining term lengths are economically significant variables not included in the current feature set.

Third, the dataset represents a static 2025–2026 price snapshot. It does not model the time dimension of price changes, limiting its use for forecasting applications.

VI. CONCLUSION

This paper presented a systematic comparison of machine learning regression methods for house price prediction in Addis Ababa, Ethiopia. Training on a realistic dataset of 500 properties across 19 neighborhoods, Gradient Boosting achieved an R^2 of 0.962 and an MAE of 2.67 million ETB, outperforming Linear Regression by 8.4 percentage points in R^2 and reducing prediction error by approximately 59%. The results demonstrate that non-linear ensemble methods are essential for capturing the multiplicative location premiums and compound condition-age interactions that characterize Ethiopian real estate pricing.

Neighborhood zone and house area emerge as the strongest predictors, consistent with hedonic price theory, but with magnitude and directionality shaped by Ethiopia-specific market dynamics including diaspora investment, infrastructure development along the ring road, and constrained supply of new high-quality housing.

Future work should focus on three directions: (1) collecting real transaction data from platforms such as Ethiopian Property Centre to validate and retrain the models; (2) incorporating additional features such as building type (villa, apartment, condominium), lease term remaining, and GPS coordinates to enable geospatial modeling; and (3) extending the framework to a time-series forecasting model as annual price data becomes available.

The full dataset, trained models, command-line predictor, and Streamlit web application are publicly available at: <https://github.com/Estifanos-Abera/house-price-predictor>

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The dataset and experimental code are fully reproducible and publicly available at: <https://github.com/Estifanos-Abera/house-price-predictor>

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